

Finite-Height Rigidity of the Riemann Zeta Zeros: A Preregistered Numerical Test Against Random-Matrix Statistics

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Abstract

We test whether two blocks of 2,000 consecutive nontrivial zeros of the Riemann zeta function, the lowest and those near the 100,000th, follow the statistics of the Gaussian Unitary Ensemble (GUE). Hypotheses and thresholds were fixed in a protocol committed before any result was generated. The zeros are 5 to 12 times closer to GUE than to independent random points in nearest-neighbour spacing and pair correlation, with strong level repulsion. However, the spacing variance is 0.148 (low block) and 0.162 (high block), below the 95% sampling band $[0.169, 0.195]$ of 150 simulated GUE samples of the same size and below the exact GUE value 0.18000: the zeros are measurably *more rigid* than finite-size GUE, and the discrepancy shrinks with height. This direction is consistent with published finite-height corrections to the Montgomery–Odlyzko law; that model is not tested here. The study is numerical and makes no claim about the Riemann Hypothesis.

1 Question and context

Montgomery [1] showed, assuming the Riemann Hypothesis, that the pair correlation of zeta zeros agrees with the GUE form $R_2(r) = 1 - (\sin \pi r / \pi r)^2$ for test functions with restricted Fourier support, and Odlyzko [2] found close numerical agreement at large height. The agreement is an asymptotic statement. We ask a bounded finite question: for two disjoint blocks of 2,000 zeros, how far are the empirical nearest-neighbour spacing law and pair correlation from the GUE predictions, compared with (a) a Poisson control, (b) the sampling scatter of genuine GUE data of the same size, and (c) each other.

2 Data and unfolding

Heights γ_n of the zeros $n = 1, \dots, 2000$ (heights 14.13 to 2515.29) and $n = 100000, \dots, 101999$ (heights 74920.83 to 76257.69) were computed with `mpmath` and frozen. Completeness was verified by Hardy’s function $Z(t)$, which changes sign at each simple critical-line zero: the sign of Z at the midpoint of every gap strictly alternates on all 1,999 gaps of both blocks, which a single missed or duplicated zero always breaks. Zeros are unfolded by $x_n = \theta(\gamma_n)/\pi + 1$ with θ the Riemann–Siegel theta function [5], so the spacings $s_n = x_{n+1} - x_n$ have mean 1.

3 Reference laws and null

The GUE nearest-neighbour spacing law is computed exactly from the sine-kernel Fredholm determinant $E(s) = \det(I - K_s)$ by Gauss–Legendre quadrature [3], with $F(s) = 1 + E'(s)$; its mean is 1.0000000 and variance 0.17999. The Poisson control has $F(s) = 1 - e^{-s}$ and $R_2 \equiv 1$. Because one zeta block is a single realization, every statistic is also computed on 150 simulated GUE samples (seed 20260920; central, semicircle-unfolded eigenvalues of independent 800×800 matrices pooled to 1,999 spacings), giving a calibrated sampling band. Kolmogorov–Smirnov (KS) values are used only as distances, never p -values, since zeta spacings are not independent.

4 Results

Statistic	Low block	High block	Exact GUE	GUE band (95%)
Spacing KS distance to GUE	0.0426	0.0255	—	≤ 0.0243
Pair-correlation RMS to GUE	0.0681	0.0606	—	≤ 0.0703
Spacing variance	0.1479	0.1618	0.1800	0.1686–0.1946
Share of spacings < 0.25	0.0080	0.0110	0.0163	0.0115–0.0225

Table 1: Measured statistics against exact GUE and the simulated GUE sampling band. Poisson distances are 0.3161 and 0.3052 (KS) and 0.3608 and 0.3489 (pair-correlation RMS) for the low and high blocks.

Preregistered hypotheses. **H1** (GUE closer than Poisson on both distances in both blocks): confirmed. **H2** (fewer than 7.37% of spacings below 0.25): confirmed (0.80%, 1.10%). **H3** (each distance inside the GUE sampling band): *not confirmed*; the pair-correlation RMS passes in both blocks, while the spacing KS distance exceeds the band (clearly in the low block, marginally in the high block) and the spacing variance lies below the band in both. **H4** (agreement improves with height on both distances): confirmed.

5 Interpretation and limits

The excess rigidity and its decrease with height are the direction expected if, at finite height, the zeros behave like a unitary ensemble of finite effective dimension growing logarithmically with height, as developed by Bogomolny, Bohigas, Leboeuf and Monastera [4]. We did not fit or test that model. Further limits: two blocks only; spacing and pair correlation only; smooth-density unfolding only; the simulated band carries its own sampling error of a few percent, so the high block’s marginal KS exceedance (0.0255 against 0.0243) should be read cautiously; and each block is one realization, so the height trend establishes direction, not magnitude. No claim is made about the Riemann Hypothesis.

6 Proposed follow-up (not a result)

Preregister a v0.2 that replaces the infinite- N GUE reference by a circular unitary ensemble of effective dimension set from the published formula without fitting, and test whether both blocks then fall inside that model’s sampling band.

Reproducibility and AI assistance

Code, frozen zeros, protocol, registry and tests:

<https://github.com/lindgreendavid/zeta-zeros-lab>

Interactive laboratory:

<https://zeta-zeros-lab-interactive.lindgreendavid.workers.dev>

The protocol commit precedes the registry commit in the repository history. The analysis code, tests and drafts of this text were produced with substantial assistance from an AI system (Claude, Anthropic); the author is responsible for the content.

References

- [1] H. L. Montgomery, The pair correlation of zeros of the zeta function, *Proc. Sympos. Pure Math.* **24** (1973), 181–193.
- [2] A. M. Odlyzko, On the distribution of spacings between zeros of the zeta function, *Math. Comp.* **48** (1987), 273–308.
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- [4] E. B. Bogomolny, O. Bohigas, P. Leboeuf, A. G. Monastera, On the spacing distribution of the Riemann zeros: corrections to the asymptotic result, *J. Phys. A* **39** (2006), 10743–10754.
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